

Validity Analysis: ARABICMMLU

Target context: MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Overall risk: **HIGH**

Deployment Context

Use case: Educational tutoring and supporting students studying with answers related to their curriculums in different subjects on the primary, middle, and high school, level as well as some university level subjects (management, economics, law, CS, political science, accounting)

Target population: Students (primary, middle, high school, and university) from Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	3	Moderate gaps	high
Input Content 	2	Significant gaps	high
Input Form 	4	Minor gaps	high
Output Ontology	3	Moderate gaps	high
Output Content 	2	Significant gaps	high
Output Form 	4	Minor gaps	high
Average	3.0		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

ArabicMMLU is the only large-scale natively-sourced Arabic exam benchmark available and is genuinely better-suited for MENA evaluation than translated MMLU alternatives [Q5, Q6]. For the deployment's MSA school-level STEM coverage and overall MCQ format alignment, it provides meaningful and trustworthy signal. However, for the deployment's eight-country curriculum-alignment requirement, ArabicMMLU has serious validity limitations that the dataset analysis confirms empirically rather than merely speculatively. The most acute concerns are content-level (IC) and label-correctness (OC): Jordanian-curriculum civics, history, social science, and economics content is encoded as universal ground truth across school

levels [DATASET-D1–D7, D20–D22, D25, D30]; university-level non-CS subjects come almost exclusively from a single Egyptian university [DATASET-D39, D40]; Law (Professional) is exclusively Moroccan procedural law in the sample [DATASET-D8–D10, D34]; Kuwait has no observed examples and no documented annotators [Q31, Q96]; Palestinian content (top-three contributor by volume) was verified without Palestinian annotators [Q96, WEB-1]. The flat label schema [Q27] cannot encode jurisdiction-specific correctness despite the deployment user explicitly requiring per-country legal tailoring [elicitation A2]. University-level coverage is also structurally thin (~6.1% [Q71]) for a deployment serving university students.

ID	Evidence
Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
D1	What are the manifestations of change in modern Jordanian society? — country-specific Jordanian civics
D2	The first coalition government in Jordan in 1952 was known as the ___ government — Jordanian political history
D3	بوهة فاك ةلودل نوؤش لىل ةماعل آلؤل بحاص سللم — The council with general sovereignty over all state affairs is the Council of Ministers — specifically Jordanian constitutional structure
D4	Rights under the [Jordanian] Kingdom's constitution — explicitly references the Jordanian Hashemite Kingdom
D5	كلمل ةلأل دهع فى 1952 ماع روتسد رص — The 1952 constitution was issued during the reign of King [Talat bin Abdullah] — Jordanian royal history
D6	وهو يزك رمل كنبل ينورتكلال لقومل قىرط نع ىوكشل لاسرا — Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo — explicitly references Jordan's central bank
D7	67% ةىسرلر ةلامل تامدخ لىل لوصولو نوعىطستسى ال نذل نىندرأل ةبسن — 67% of Jordanians cannot access formal financial services — Jordan-specific statistic as universal economic fact
D8	ىوعدل رهوچ فى مكحل رودص دعب الل عوفف وأ ضراع غازن فى ةرداصل وأ ةىدمهممتل ماكحلأل فانئتسا لبقى ال — Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D9	عافدلابل ةلكلمل ةىموكحل ةطلسلو لدعل ربزو نم رارقب...نىكردلل ةئاضقل ططرشل طباض ةفص لىوخت نكمى — Moroccan judicial police powers — references "الملكى كردل" (Royal Gendarmerie of Morocco)
D10	محى حص ةبوجل عىم: مكحل رودص لبقى ندىم لفرطل لزانن اذ — Moroccan civil party withdrawal in criminal proceedings
D20	Royal Medical Services — ةىبطل ةنىدمل ىفشستسم: انهم...تاسسؤم نم دىدل ةىموكحل ةىلكلمل ةىبطل تامدخ لمشت — Medical City Hospital — specifically Jordanian Royal Medical Services
D21	King Talal bin Hussein's reign ended for health reasons in 1952 — one year — Jordanian royal history in primary school
D22	Lady of the Mountain Church is in Anjara [Jordan] — ةرجنع فى لبلجل ةدىس ةسىكن دجوت — Jordanian religious geography
D25	The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions — Jordanian national history
D30	Wadi Rum is a Jordanian nature reserve under joint management — مرى داو...كترتشم ةرادل اهلل فرشت ندرأل فى ةىبطل ةىمحم — Jordan-specific geography fact
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF

D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)

Practical Guidance

What This Benchmark Measures

ArabicMMLU can validly measure: (a) MSA reading comprehension and Arabic grammar at school level (input_form score 4); (b) MCQ-format scoring alignment for the deployment's primary use case (output_form score 4); (c) STEM factual recall across all eight countries since STEM content is curriculum-neutral [DATASET-D24]; (d) Jordanian school-curriculum knowledge specifically (the dominant cohort); (e) some Egyptian university accounting/economics/management/political science alignment, subject to OCR caveats; (f) Moroccan procedural law specifically. The benchmark also supports rough comparative evaluation of model robustness across prompt/output configurations [Q47, Q105–Q110].

Construct Depth

Construct depth is shallow for the deployment's stated needs. The benchmark probes factual recall and modest reasoning at primary/middle/high school levels well, but university-level depth is thin (~6.1% of items, ~889 questions across all subjects [Q71]). Country-specific construct depth is highly uneven: Jordan is over-probed, Kuwait is essentially un-probed, Morocco is probed only in law and a few Arabic items, Palestine is probed in STEM but not verified by Palestinian experts. Madhab-specific Islamic studies depth is undocumented [gap_id 7]. Jurisdiction-specific legal depth is single-jurisdiction in the sample [DATASET-D34]. Construct depth strengths are concentrated in dimensions input_form and output_form; depth weaknesses are concentrated in input_content and output_content — the dimensions the elicitation marked HIGH priority.

What Else You Need

For a defensible deployment, ArabicMMLU should be supplemented with: (1) per-country re-annotation of civics, history, social science, economics, and law subsets by curriculum experts from Morocco, Kuwait, Palestine, UAE (addresses input_content and output_content gaps); (2) ArabLegalEval [WEB-3] for Saudi-specific law and MizanQA [WEB-4] for additional Moroccan law coverage, with a custom evaluation built for Egyptian/UAE/Jordanian/Lebanese/Kuwaiti/Palestinian legal jurisdictions which remain unfilled; (3) AraSTEM [WEB-2] for richer STEM coverage at university level (addresses input_ontology thinness); (4) a French-Arabic bilingual evaluation for Moroccan university science (no existing benchmark, must be built [gap_id 6]); (5) madhab-stratified Islamic studies evaluation built with Maliki and Hanbali curriculum experts [gap_id 7]; (6) cleaning or replacement of OCR-corrupted Management and Political Science (University) items [DATASET-D14, D15, D29].

ID	Evidence
Q47	"We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output." (p.5)
Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
Q105	"Zero-shot LLM performance (% accuracy) with Arabic prompt and Arabic alphabetic output, combined across subject groups." (p.16)
Q106	"Average" means the average across all questions in ArabicMMLU." (p.16)
Q107	"Table 9 presents results organized by subject groups: STEM, Social Science, Humanities, Arabic Language, and Other." (p.16)
Q108	"Zero-shot LLM performance (% accuracy) with Arabic prompt and English alphabetic output, combined across subject groups." (p.17)
Q109	"Table 10: Zero-shot LLM performance (% accuracy) with Arabic prompt and English alphabetic output, combined across subject groups." (p.17)

Q110	“Zero-shot LLM performance (% accuracy) with English prompt and Arabic alphabetic output, combined across subject groups.” (p.18)
D14	Severely OCR-distorted text about authority distribution in organizations, from Egyptian source — د14 — Severe OCR-distorted text about authority distribution in organizations, from Egyptian source
D15	Severely OCR-corrupted management question with unreadable answer options — د15 — Severe OCR-corrupted management question with unreadable answer options
D24	Calvin cycle question — د24 — Calvin cycle question — Palestine-sourced biology; universal factual content
D29	Severely OCR-distorted Arabic text about conditions for an Islamic ruler — from Egyptian source — د29 — Severe OCR-distorted Arabic text about conditions for an Islamic ruler — from Egyptian source
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF — د34
WEB-2	https://arxiv.org/abs/2501.00559
WEB-3	ArabLegalEval reuses ArabicMMLU law subsets without jurisdiction tagging — confirms multi-jurisdiction gap is structural (https://arxiv.org/abs/2408.07983) — WEB-3
WEB-4	Arabic LLM benchmark survey confirms no French-Arabic bilingual benchmark for Moroccan higher education exists (https://arxiv.org/abs/2510.13430) — WEB-4

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

ArabicMMLU's taxonomy of 40 tasks across STEM, Social Science, Humanities, Arabic Language, and Other [Q25, Q107], spanning primary through university levels, broadly matches the deployment's educational tiers and subject scope. All six university subjects required by the deployment (law, CS, management, economics, political science, accounting) are confirmed present in the benchmark configurations [DATASET-D8, D11, D13, D14, D16]. However, university-level questions constitute only ~6.1% of the benchmark [Q71], yielding roughly ~889 questions spread across all university subjects — thin coverage for a deployment that explicitly targets university students. Country coverage at the ontology level is structurally uneven: Jordan dominates with over 6,000 questions while some countries contributed 'as few as 100 or, in some cases, none at all' [Q89]. Several configurations have null country/level metadata [DATASET-D26, D27], limiting their use for country-specific diagnostics. The taxonomy includes no madhab labeling for Islamic studies and no jurisdiction-specific branching for law, both flagged as HIGH-priority deployment requirements.

ID	Evidence
Q25	"ArabicMMLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels." (p.3)
Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
Q107	"Table 9 presents results organized by subject groups: STEM, Social Science, Humanities, Arabic Language, and Other." (p.16)
D8	— Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure يؤدى لدل رهوج يف مكحل رودص دعب الإ عوفد وأ ضراع عازن يف ةرداصل وأ ةيديهمم التل ماكحلأ فانئتسا لبقي ال
D11	— "Sequential events, cooperative or conflictual, are called: interactions" — general IR concept تالعات التل: هى ةىعارص وأ ةىنواع نوك دق ةعاتم تادحا
D13	— Computer auditing concept — general accounting principle, from Egyptian source ةىعبتلاب ةمىمس نوك لىغشتل ةىممع نإف...ةمىمس تلأخدم تنك اذا ونأ بساحل لوح ةعجارمل بومسأ ضرثفى
D14	— Severely OCR-distorted text about authority distribution in organizations, from Egyptian source لىل ري تذت...ةتفلخسل ةتيرادلإ ةاىهتدسل او صاختلأل ني تب ةطلتدل عت يهت ةتجرد
D16	— "A microcomputer is a desktop computer" — from KSA source (faculty.ksu.edu.sa) ال — بيكتلم رتوي بمكل زاهج: ن عرابع وه قيقدل رتوي بمكل
D26	All 5 examples: Country = null, Source = folderat.com, Subject = Islamic Studies — Islamic Studies "All" config examples have no country metadata — source appears to be generic Arabic quiz site
D27	All 5 examples: Country = null, Source = madinaharabic.com — Arabic Language General has no country metadata — sourced from online Arabic language learning site

Strengths

- All six deployment-required university subjects are present as queryable configurations [DATASET-D8, D11, D13, D14, D16]

- Multi-level coverage (primary, middle, high school, university) is empirically confirmed in the data [DATASET-D5, D13, D21]
- Subject taxonomy explicitly includes country-variation-prone subjects: history, geography, civics, political science [Q72, Q73]
- Negation is treated as a distinct linguistic phenomenon [Q83, Q84], providing some construct depth beyond factual recall

ID	Evidence
Q72	"We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts." (p.7)
Q73	"These subjects include history, geography, civics, political" (p.7)
Q83	"Despite negation being an absolutely foundational linguistic phenomenon, LLMs have been shown to be worryingly insensitive to its effects in English (Kassner and Schütze, 2020; Hosseini et al., 2021; Truong et al., 2023)." (p.8)
Q84	"We thus perform an analysis of LLM performance over questions in ArabicMMLU with and without negation to determine whether this observation ports across to Arabic." (p.8)
D5	"كلمة الملك عبد الله الأول ملك الأردن في 1952 مع روتس د رص" with options referring to Jordanian kings — "The 1952 constitution was issued during the reign of King [Talal bin Abdullah]" — Jordanian royal history
D8	— يوعدل رهوج في مكحل رودص دعب الإ عوفد وأ ضراع عازن في أرداصلا وأ ديديهمم التل ماكحلأ فانئتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D11	— "Sequential events, cooperative or conflictual, are called: interactions" — general IR concept
D13	— عيبتل اب قمي مس نوكت لي غشتل ايمم نإف... قمي مس تل اخدم تل تنك اذ ونأ بساحل ل و ح ع ج ا ر م ل ب و م س أ ض ر ت ف ي Computer auditing concept — general accounting principle, from Egyptian source
D14	— ل ي ل ي ت ذ ت... ق ت ف ل ت خ س ل ل ق ت ي ر ا د ل إ ل ت ا ي ه ت د س ل ل و ص ا خ ل ت أ ل أ ن ي ت ب ط ل ت د ل ع ت ي ي ه ت ق ت ج ر د OCR-distorted text about authority distribution in organizations, from Egyptian source
D16	— ي ب ك م ل ر ت و ي ب ك م ل ز ا ه ج : ن ع ق ر ا ب و ه ق ي ق د ل ر ت و ي ب ك م ل "A microcomputer is a desktop computer" — from KSA source (faculty.ksu.edu.sa)
D21	— ق ن س ه م ك ح د م و م 1952: ي ف ك ل ذ و ق ح ص ب ا ب س أ ل ه ت ي ا ل و ت ه ت ن ا ن ي س ج ل ن ب ل ل ط ل ل م ل "King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories include MSA-medium primary/middle/high school content for Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, KSA across STEM, humanities, civics, Islamic studies, Arabic language; plus university-level coverage for law, CS, management, economics, political science, accounting [elicitation A1, A2].

IO-2: Several regionally-relevant categories are omitted or thinly covered: madhab-specific Islamic studies tracks (Maliki for Morocco, Hanbali for KSA) have no labeling [gap_id 7]; jurisdiction-specific law branches are absent (flat label schema [Q27]); Kuwaiti civic/national content appears absent — no Kuwaiti workers documented [Q31, Q96] and no Kuwaiti examples observed in data sample [DATASET content summary]; French-medium Moroccan university science content is fully out of scope [Q92].

IO-3: Some included categories are not directly relevant to all deployment cohorts: 'Driving Test' configuration [DATASET-D35] is not a tutoring subject; null-country generic Islamic Studies and Arabic Language items [DATASET-D26, D27] cannot be used for country-stratified evaluation. These are not harmful inclusions per se but contribute construct-irrelevant variance to

aggregate scores.

IO-4: Documented gaps: (a) university-level coverage at ~6.1% [Q71] is structurally thin for a deployment serving university students; (b) no madhab calibration [gap_id 7]; (c) no jurisdiction-specific law taxonomy [Q27, gap_id 4]; (d) Kuwait coverage likely near-zero [Q89, Q96]; (e) French-medium Moroccan content fully absent [Q92].

ID	Evidence
Q25	"ArabicMMLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels." (p.3)
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
Q107	"Table 9 presents results organized by subject groups: STEM, Social Science, Humanities, Arabic Language, and Other." (p.16)
D26	All 5 examples: Country = null, Source = folderat.com, Subject = Islamic Studies — Islamic Studies "All" config examples have no country metadata — source appears to be generic Arabic quiz site
D27	All 5 examples: Country = null, Source = madinaharabic.com — Arabic Language General has no country metadata — sourced from online Arabic language learning site
D35	رتم 50 دعب ىلع؟ ععضو كىل ع نيأ، سكاغلا ريذحتلا ثلثم كىدلو كتبكرم لاطعت لاج يف — UAE driving rule (50m warning triangle placement) — country-specific traffic regulation
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)
WEB-4	Arabic LLM benchmark survey confirms no French-Arabic bilingual benchmark for Moroccan higher education exists (https://arxiv.org/abs/2510.13430)

Information Gaps

- Exact per-country question counts beyond Jordan (>6,000) are not text-published [gap_id 5]
- Madhab calibration of Islamic studies questions is undocumented anywhere [gap_id 7]
- Jurisdictional distribution of law questions beyond Moroccan procedural law sample is unknown

Requires Expert Verification

- Whether observed Moroccan-only Law (Professional) sample generalizes to the full ~299 law questions or whether other configurations contain embedded jurisdiction-specific content
- Whether Islamic Studies configurations contain madhab-specific fiqh that would diverge across Maliki/Hanbali/Shafi'i traditions

Remediation

Gap 1: University-level coverage is ~6.1% of total [Q71] (~889 questions across all subjects), and Law (Professional) is exclusively Moroccan in the sample [DATASET-D34]; jurisdiction-specific tagging is absent.

Recommendation: Supplement with ArabLegalEval [WEB-3] for Saudi law, MizanQA [WEB-4] for Moroccan law, and AraSTEM [WEB-2] for STEM university content. Commission custom evaluation sets for Egyptian, UAE, Jordanian, Lebanese, Kuwaiti, and Palestinian legal jurisdictions which remain entirely unfilled by existing Arabic benchmarks.

ID	Evidence
----	----------

Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
WEB-2	https://arxiv.org/abs/2501.00559
WEB-3	ArabLegalEval reuses ArabicMMLU law subsets without jurisdiction tagging — confirms multi-jurisdiction gap is structural (https://arxiv.org/abs/2408.07983)
WEB-4	Arabic LLM benchmark survey confirms no French-Arabic bilingual benchmark for Moroccan higher education exists (https://arxiv.org/abs/2510.13430)

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Content-level validity is the dimension where ArabicMMLU diverges most sharply from the deployment's eight-country requirement. While the benchmark's content is sourced from real native Arabic exams [Q1, Q26] — a clear strength over translated alternatives [Q5, Q6] — the empirical content distribution is heavily skewed: civics, social science, history, and economics at school level are dominated by Jordanian-specific institutional, royal, and constitutional content presented as universal curriculum knowledge [DATASET-D1, D2, D3, D4, D5, D6, D7, D20, D21, D22, D25, D30, D31]. University-level economics, accounting, management, and political science come exclusively from a single Egyptian university (Assiut University) in the sample [DATASET-D13, D14, D28, D39, D40], and Law (Professional) is exclusively Moroccan in the observed sample [DATASET-D8, D9, D10, D34]. The deployment user explicitly requires country-specific tailoring for law [elicitation A2] and curriculum-aligned content for each of the eight countries [elicitation A1]. Kuwait has no observed examples and no documented annotators [Q31, Q96]; Morocco has no documented annotators [Q31, Q96]. Egypt-specific currency (ج.م.ع) appears in university economics presented as universal [DATASET-D28]. The benchmark's own admission that countries are represented unequally [Q89] is empirically confirmed and amplified by the dataset analysis: rather than uniform under-coverage, the data shows Jordanian-curriculum content is propagated as default for non-Jordanian cohorts. This generates substantial construct-irrelevant variance for Moroccan, Emirati, Kuwaiti, Saudi, Lebanese, and Palestinian students.

ID	Evidence
Q1	"we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions." (p.1)
Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)
Q26	"The questions are sourced from eight different countries in North Africa (Morocco and Egypt), the Levant (Jordan, Palestine, and Lebanon), and the Gulf (UAE, Kuwait, and KSA)." (p.3)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
D1	What are the manifestations of change in modern Jordanian society? — country-specific Jordanian civics
D2	The first coalition government in Jordan in 1952 was known as the ___ government — Jordanian political history
D3	The council with general sovereignty over all state affairs is the Council of Ministers — specifically Jordanian constitutional structure
D4	Rights under the [Jordanian] Kingdom's constitution — explicitly references the Jordanian Hashemite Kingdom

D5	"لكلمة الإلحاح دعه في 1952 ماع روتسد رص" with options referring to Jordanian kings — "The 1952 constitution was issued during the reign of King [Talal bin Abdullah]" — Jordanian royal history
D6	وهو يزكركم كنبلل ينورتكلال عقومل قيرط نع يوكشلال لاسرا www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank
D7	0.67: نسبة من الأردنيين لا يستطيعون الوصول إلى الخدمات المالية الرسمية — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact
D8	— يوعدل رهوج في مكحل رودص دعب الإ عوفد وأ ضراع عازن في مرداصل وأ عيدي مهممتل ماكحلأ فائتتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D9	عافدل اب فلكملم عيوكحل اطلسل او لدعل ربزو نم رارب...نيكي كدلل عيئاضلق اطرشل طباض ففص لي وخت نكمي "الملك لردل" (Royal Gendarmerie of Morocco) — Moroccan judicial police powers — references
D10	— Moroccan civil party withdrawal in criminal proceedings — عحي حص بوقلأ عي مج: مكحل رودص لبقي ندمل فرطل لزانن اذإ
D13	— عي عبتال ب عيمس نوكت لي غشتل ا عي ممع نإف... عيمس تال خدمل تنك اذإ ونأ بساحل لوح ععجارمل بومسأ ضررتفي Computer auditing concept — general accounting principle, from Egyptian source
D14	Severely — لي ري تذت... فتتلفل خسل ا تي رادل ا تاي هتتدسل او صاختل ألي ني تب عطلتدل عتي هت فتجرد OCR-distorted text about authority distribution in organizations, from Egyptian source
D20	"Royal Medical Services include: Medical City Hospital" — عي بطل ا ن عي دمل ا ي فشتسم: اهنم... تاسس ؤمل ا نم دي دعل ا عي وكحل ا عي كمللم ا عي بطل ا تادمخل ا لمشت Royal Medical Services specifically Jordanian Royal Medical Services
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — عنس همك ح دموم، 1952: في كل ذو عي حص بابسأل هتالو تهتن ا ن عي سحل ا نب لال ط لكلمل Jordanian royal history in primary school
D22	— Lady of the Mountain Church is in Anjara [Jordan] — عرجنع في لبجل ا عديس ا عسي نك دجوت Jordanian religious geography
D25	"The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — ماع هي ناطي ربل ا هي ندرأال ا هدا عمل ل سيئرل ا فدهل ا نك Jordanian national history
D28	— Egyptian currency (ج = جنيه) ١٠ % : وه عرمثتسمل ا ل اومال ا لي ع دئاعل ا لدعم... هي نك ١٠٠٠٠ بولطم ا رامثتسال ا مجح نك اذإ Egypt-specific monetary context
D30	"Wadi Rum is a Jordanian nature reserve under joint management" — مري داو... كعرتشم ا راد ا اهل ا فرشت ندرأال ا في عي عي بط عي مجح Jordan-specific geography fact
D31	King Abdullah I's positions on Zionism — ني طسلف في اعا ماطأ و عي نو يهصل ا كرحل ا نم لوال ا هل لادبع لكلمل ا فقاوم نم تسلي ا عي تال ا نم دحاو highly sensitive political content, Jordanian framing of Palestinian history
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source

Strengths

- Native exam provenance avoids translation artifacts that plague competing Arabic benchmarks [Q5, Q6]
- STEM content (math, biology, physics, CS factual) is genuinely curriculum-neutral and reliable across all eight countries [DATASET-D24]
- Palestine is substantively represented in STEM subjects (Biology HS, CS Primary, Math Primary) [DATASET-D23, D24]
- Some Moroccan legal content is present, even if narrow [DATASET-D8, D9, D10]
- Native annotators and exam URLs document content provenance per question [Q32, Q33]

ID	Evidence
Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)

Q32	"During the first stage of data collection, the internal workers were tasked with collecting relevant sources for data collection. These sources were URLs containing the questions, which needed to be publicly available." (p.4)
Q33	"In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4)
D8	— Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure يؤعدل رهوج يف مكحل رودص دعب الإ عوفد وأ ضراع عازن يف ةرداصل وأ أيديهم التامك أال فانئتسا لبقي ال
D9	عافدل اب ةفل كم لم ةيموكحل اطلسل او لدعل ربزو نم رارقب...نني كيدل ةيئاضق ل طرشل طباض ةفص لي وخت نكمي Moroccan judicial police powers — references "لكم ل كيدل" (Royal Gendarmerie of Morocco)
D10	Moroccan civil party withdrawal in criminal proceedings — عحي حص ةب و أال عي م ح : مكحل رودص لبقي ندمل فرطل ل زانت اذا
D23	"The binary number system is used by computers" — Palestine-sourced primary CS دأ — ثنائيل ماطنل : اهتجلاعمو تانايبل طفحو لي ثمت يف بوساحل همدختسي دعل ةمطنأ دأ
D24	Calvin cycle question — Palestine-sourced biology; universal factual content إذا — Calvin cycle question — زوكولغ تائي زج (3) جتن اذا

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Inputs heavily require region-specific knowledge but the regions covered diverge from the deployment's eight-country target. Civics and history items reference Jordanian royal lineage, the 1952 Jordanian constitution, Jordan's Royal Medical Services, and Central Bank of Jordan URLs as universal facts [DATASET-D1, D2, D4, D5, D6, D20, D21, D25, D31]. University accounting and economics use Egyptian institutional and currency context [DATASET-D28, D39, D40]. Law uses Moroccan procedural code [DATASET-D8, D9, D10].

IC-2: Cultural-content alignment is partial. Islamic studies content observed (Hadith ethics, Quran recitation etiquette) appears broadly Sunni-consistent, aligning with the deployment's predominantly Sunni population [DATASET-D17, D37, D38]. However, no madhab calibration exists [gap_id 7], and Hanbali/Maliki specificity is not verified. Politically sensitive content (e.g., King Abdullah I and Zionism [DATASET-D31]) is framed in Jordanian national narrative — potentially misaligned with Palestinian, Lebanese, or other framings.

IC-3: Western-specific knowledge is largely avoided due to native exam sourcing [Q5, Q6]. The primary content-validity threat is not Western bias but intra-regional bias — Jordanian and Egyptian content dominating non-Jordanian/non-Egyptian cohorts.

IC-4: Annotators are exclusively from Jordan, Egypt, Lebanon, UAE, KSA [Q31] — Morocco, Kuwait, Palestine have no documented annotators despite (in Palestine's case) contributing top-three question volume [Q96, WEB-1]. Recruiting Kuwaiti, Moroccan, and Palestinian curriculum experts would be essential for any deployment use.

IC-5: Documented content issues harming validity: (a) Jordanian-default civics/history/social science presented as universal [DATASET-D1, D2, D4, D21]; (b) single-Egyptian-university dominance of university social science subjects [DATASET-D39, D40]; (c) single-source single-jurisdiction Law (Professional) [DATASET-D34]; (d) no Kuwait content observed; (e) Egypt-specific currency in 'general' economics [DATASET-D28]; (f) French-medium Moroccan university content absent [Q92].

ID	Evidence
----	----------

Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
D1	"What are the manifestations of change in modern Jordanian society?" — country-specific Jordanian civics
D2	"The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D4	"Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom
D5	"The 1952 constitution was issued during the reign of King [Talal bin Abdullah]" — Jordanian royal history
D6	www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank
D8	appeals rules under Moroccan Code of Civil Procedure
D9	references "Royal Gendarmerie of Morocco" (Moroccan judicial police powers)
D10	Moroccan civil party withdrawal in criminal proceedings
D17	Country: Jordan — 5-option Islamic studies question about positive effects of thinking
D20	"Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school
D25	"The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history
D28	Egyptian currency (Egyptian pound) in university economics — Egypt-specific monetary context
D31	King Abdullah I's positions on Zionism — highly sensitive political content, Jordanian framing of Palestinian history
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D37	Country: Jordan — Generic Islamic etiquette of Quran recitation — no madhab specificity indicated
D38	Country: Jordan — Islamic studies question about examples in Quran — Jordanian curriculum framing
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)

Information Gaps

- Whether Jordanian dominance in civics/history/social science extends to the full benchmark or is concentrated in specific level/subject configurations [gap_id 5]
- Whether Law subset beyond the sampled Moroccan questions includes other jurisdictions [gap_id 4]
- Madhab-specific framing in Islamic studies is undocumented [gap_id 7]

Requires Expert Verification

- Per-country curriculum review by national-curriculum experts from Kuwait, Morocco, Palestine, UAE to identify questions presented as 'universal' that are actually national-curriculum-specific
- Islamic studies curriculum experts to assess madhab calibration risk for Moroccan Maliki and Saudi Hanbali contexts

Remediation

Gap 1: Jordanian-curriculum civics, history, social science, and high-school economics content is encoded as universal across configurations, creating construct-irrelevant variance for the seven non-Jordanian deployment cohorts [DATASET-D1, D2, D6, D7, D21, D25].

Recommendation: Use the per-question country metadata field [Q33] to filter or stratify evaluation by source country; do not aggregate across countries for civics, history, or social science. Build a Kuwaiti, Moroccan, and Palestinian curriculum supplement before deploying to those cohorts.

Gap 2: Islamic studies has no madhab labeling anywhere in the benchmark [gap_id 7]; observed Islamic Studies HS content is Jordanian-curriculum-sourced [DATASET-D38] without Maliki (Morocco) or Hanbali (KSA) calibration.

Recommendation: Engage Islamic studies curriculum experts from Morocco (Maliki), KSA (Hanbali), and a Shafi'i-region representative to audit a stratified sample of Islamic studies items and tag madhab-specific items; flag madhab-divergent questions for country-conditioned evaluation.

ID	Evidence
Q33	"In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4)
D1	"What are the manifestations of change in modern Jordanian society?" — country-specific Jordanian civics
D2	"The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D6	www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank
D7	0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school
D25	"The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history
D38	"Country: Jordan — Islamic studies question about examples in Quran — Jordanian curriculum framing

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form aligns well with the deployment. All 14,575 questions are MSA [\[Q2\]](#), matching the formal medium of instruction across all eight target countries [\[Q29\]](#). The text-only, MCQ format with embedded contextual passages [\[Q37, Q38\]](#) is exactly the deployment's primary use case. Arabic-script fidelity is empirically confirmed [\[DATASET-D24, D25\]](#). The MSA-only scope appropriately excludes Darija (which the deployment explicitly excludes) and dialectal Arabic [\[Q92\]](#). The main form-level gap is French-medium Moroccan university content [\[Q92\]](#), which the deployment user explicitly identified as in-scope for Moroccan university science programs [elicitation A4]. A secondary form concern is OCR corruption observed in Management and Political Science (University) questions from Egyptian PDF sources [\[DATASET-D14, D15, D29\]](#), which compromises the input signal quality for those configurations. A minor anomaly is one Arabic Grammar question with English-language stem [\[DATASET-D19\]](#).

ID	Evidence
Q2	"Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region." (p.1)
Q29	"In public schools, Arabic is commonly used for teaching and assessment, while in international schools, English is the predominant language of instruction for most subjects, following either the UK or USA curriculum." (p.3)
Q37	"During manual data scraping, workers were instructed to include only questions accompanied by an answer key, and to discard questions containing multi-modal information (e.g., images, videos, or tables)." (p.4)
Q38	"If a question had additional contextual information (e.g., a passage referenced by several questions), the context was required to be included with each corresponding question." (p.4)
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
D14	Severely OCR-distorted text about authority distribution in organizations, from Egyptian source إلى ري ت ذت...ة ف ت خ س ل ا ت ي ر ا د ل ا ا ت ا ي ه ت ت د س ل ا و ص ا خ ت ل ا ل ا ن ي ت ب ة ط ل ت د ل ا ع ت ي ي ه ت ة ت ج ر د
D15	Severely OCR-corrupted management question with unreadable answer options ت ح ل ا ... ت م ح ر خ ر ل ا ... م ا د خ ت س ل ا ا ت س ن ا د ط ط خ ل ا ي ل ع ل ا ا ث م د ع ي ل ا
D19	In the following Quranic verse, what is the correct parsing of the word كَد — English-language question stem in an Arabic grammar dataset
D24	Calvin cycle question — ا ه ت ي ب ث ت م ت ي ت ا ل ا (CO2) ت ا ئ ي ز ج د د ع م ك ف ن ف ل ا ك ة ق ل ح ن ع ز و ك و ل غ ت ا ئ ي ز ج (3) ج ت ن ا ذ ا Palestine-sourced biology; universal factual content
D25	"The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history ب ا د ت ن ا ل ا ك ف د و ي ق ن م د ا ل ب ل ر ر ح ت : ١٩٢٨ م ا ع ه ي ن ا ط ي ر ب ل ا ه ي ن د ر ا ل ا ه د ا ع م ل ل س ي ئ ل ا ف د ه ل ا ن ا ك
D29	Severely OCR-distorted Arabic text about conditions for an Islamic ruler — from Egyptian source م ل ا ر ر س ل ا ر ر ل و د ل ا ر ر ف م ك ا ر ر ل ا ر ر ف ا ر ر ف ا و ت ب ر ر ر ر ت ل ا ر ر س ل ا ل ا ط و ر ر ر ش ل ا ن ر ر م

Strengths

- MSA throughout matches the formal medium of instruction in all eight countries [\[Q2, Q29\]](#)
- MCQ + 2–5 options format directly matches deployment's primary use case [\[Q27\]](#)
- Text-only modality matches the deployment's text-based interface [\[Q93, infrastructure section\]](#)

- Contextual passages bundled with dependent questions [Q38] preserves single-shot evaluability

ID	Evidence
Q2	“Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region.” (p.1)
Q27	“Each question has 2–5 candidate answers, with one correct answer.” (p.3)
Q29	“In public schools, Arabic is commonly used for teaching and assessment, while in international schools, English is the predominant language of instruction for most subjects, following either the UK or USA curriculum.” (p.3)
Q38	“If a question had additional contextual information (e.g., a passage referenced by several questions), the context was required to be included with each corresponding question.” (p.4)
Q93	“ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work.” (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions align: MSA Arabic text input matches the deployment's text-based interface and matches the formal language of instruction across all eight countries [Q2, Q29]. No image/audio mismatch since both the benchmark and deployment exclude multimodal content [Q37, Q93].

IF-2: Regional infrastructure supports text-based delivery: ITU 2024 data shows ~100% internet penetration in UAE/KSA/Kuwait [WEB-8], ~83-88% in Levant per Arab Barometer [WEB-10], and majority mobile-dominant access across all eight [WEB-11]. Mobile-dominant access is compatible with text MCQ delivery.

IF-3: Domain-specific form gap: French-medium Moroccan university science content is entirely outside benchmark scope [Q92]; deployment explicitly identifies this as in-scope for Moroccan universities [elicitation A4]. No companion benchmark identified [gap_id 6]. Additionally, OCR corruption in some Egyptian-sourced university PDFs [DATASET-D14, D15, D29] degrades input form for those subjects.

IF-4: Documented form mismatches: (a) French-medium Moroccan university content absent [Q92, gap_id 6]; (b) OCR-corrupted text in Management and Political Science university questions [DATASET-D14, D15, D29]; (c) one English-stem grammar question [DATASET-D19]; (d) several configurations have null country/level metadata limiting form-level diagnostics [DATASET-D26, D27].

ID	Evidence
Q2	"Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region." (p.1)
Q29	"In public schools, Arabic is commonly used for teaching and assessment, while in international schools, English is the predominant language of instruction for most subjects, following either the UK or USA curriculum." (p.3)
Q37	"During manual data scraping, workers were instructed to include only questions accompanied by an answer key, and to discard questions containing multi-modal information (e.g., images, videos, or tables)." (p.4)
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
Q93	"ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work." (p.9)
D14	Severely OCR-distorted text about authority distribution in organizations, from Egyptian source

D15	ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... — Severely OCR-corrupted management question with unreadable answer options
D19	In the following Quranic verse, what is the correct parsing of the word كَب — English-language question stem in an Arabic grammar dataset
D26	All 5 examples: Country = null, Source = folderat.com, Subject = Islamic Studies — Islamic Studies "All" config examples have no country metadata — source appears to be generic Arabic quiz site
D27	All 5 examples: Country = null, Source = madinaharabic.com — Arabic Language General has no country metadata — sourced from online Arabic language learning site
D29	م*لاررسالا ررلودلأ ررف مكارر لأ ررف ارر رفاوت برر* ررتلأ ررساسالا طوررررشال نررم — Severely OCR-distorted Arabic text about conditions for an Islamic ruler — from Egyptian source
WEB-4	Arabic LLM benchmark survey confirms no French-Arabic bilingual benchmark for Moroccan higher education exists (https://arxiv.org/abs/2510.13430)
WEB-8	~100% internet penetration in UAE/KSA/Kuwait — text-based output reaches target users (https://statisticsoftheworld.com/ranking/internet-users)
WEB-10	Arab Barometer 2020: ~84% Jordanian, ~83% Palestinian, ~88% Lebanese internet usage (https://www.arabbarometer.org/2020/09/the-mena-digital-divide/)
WEB-11	83% mobile ownership in Arab States; mobile-dominant text delivery is feasible (https://www.itu.int/dms_pub/itu-d/opb/ind/D-IND-SDDT-ARB-2025-PDF-E.pdf)

Information Gaps

- Total proportion of OCR-corrupted questions in Management and Political Science (University) configurations is unknown [DATASET limitations §6]

Remediation

Gap 1: OCR-corrupted text in Management and Political Science (University) configurations [DATASET-D14, D15, D29] compromises input signal quality, and French-medium Moroccan university content is entirely outside scope [Q92].

Recommendation: Quarantine or replace OCR-corrupted university management and political science items. For Moroccan university science, build or commission a French-Arabic bilingual evaluation since no existing benchmark addresses this gap [gap_id 6, WEB-4].

ID	Evidence
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
D14	ى لى لى ريتذت...ةتف ل تخس ل اة تي رادل ا تاي هت تدس ل او صا خت ل أ ل ني ت ب ة ط ل ت د ل ا ع ت ي ي ه ت ة ت ج ر د — Severely OCR-distorted text about authority distribution in organizations, from Egyptian source
D15	ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... ت. ح... — Severely OCR-corrupted management question with unreadable answer options
D29	م*لاررسالا ررلودلأ ررف مكارر لأ ررف ارر رفاوت برر* ررتلأ ررساسالا طوررررشال نررم — Severely OCR-distorted Arabic text about conditions for an Islamic ruler — from Egyptian source
WEB-4	Arabic LLM benchmark survey confirms no French-Arabic bilingual benchmark for Moroccan higher education exists (https://arxiv.org/abs/2510.13430)

Output Ontology (OO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: ARABICMMLU | Context: MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The flat single-correct-answer MCQ schema with 2–5 options [Q27] structurally matches the deployment's MCQ tutoring use case, and the user explicitly noted that for MCQ tasks explanation format is not exam-critical [elicitation A3]. This is a meaningful structural alignment. However, the schema provides no mechanism for jurisdiction-specific or country-specific correct answers. The deployment user explicitly stated 'the system must tailor legal answers to each country's specific legal system' [elicitation A2], but the benchmark's flat label structure encodes a single ground-truth answer per question [Q27], with no country-stratified branching. The dataset analysis empirically confirms this: a Moroccan procedural-law answer is simply marked correct without any 'correct-for-Morocco' qualifier [DATASET-D8, D9, D10]. For civics, history, and law subjects where correct answers legitimately vary by country, this schema cannot represent the variation the deployment requires. The benchmark does provide country, subject, and level metadata fields per question [Q33], which would in principle allow country-stratified scoring — but this is a metadata affordance rather than a label-schema accommodation, and the metadata is null for some configurations [DATASET-D26, D27].

ID	Evidence
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q33	"In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4)
D8	— يوعدل رهوج يف مكحل رودص دعب ال عوفد وأ ضراع عازن يف ةرداصل وأ أديهمتمل مكحلأل فانئتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D9	— عافدل اب ةفل كملم ةيموكحل ةطلسل او لدعل ربزو نم رارقب...نيكي كرددلل ةيئاضقلا ةطرشل طباض ةفص لي وخت نكمي Moroccan judicial police powers — references "يكي لم لك ردلا" (Royal Gendarmerie of Morocco)
D10	— ةحي حص ةبوجلأ عي مج: مكحل رودص لبقي ندمل فرطل ل زانت اذا Moroccan civil party withdrawal in criminal proceedings
D26	All 5 examples: Country = null, Source = folderat.com, Subject = Islamic Studies — Islamic Studies "All" config examples have no country metadata — source appears to be generic Arabic quiz site
D27	All 5 examples: Country = null, Source = madinaharabic.com — Arabic Language General has no country metadata — sourced from online Arabic language learning site

Strengths

- Flat MCQ schema with 2–5 options [Q27] directly matches deployment's MCQ tutoring format
- Country, subject, and level metadata fields are present per question [Q33], enabling country-stratified analysis even without label-schema branching
- Subject-group taxonomy (STEM, Social Science, Humanities, Arabic Language, Other) [Q107] supports diagnostic per-subject reporting
- Education-level stratification (primary, middle, high, university) [Q102, Q103] aligns with the deployment's tier structure

ID	Evidence
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)

Q33	"In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4)
Q102	"Table 8 presents the detailed zero-shot results across subjects and education levels, while Table 9, Table 10, Table 11 display the results with different prompts and alphabetic outputs (complementing the main result at Table 8)." (p.15)
Q103	"Zero-shot LLM performance (% accuracy) with English prompt and English alphabetic output, for each subject and education level." (p.15)
Q107	"Table 9 presents results organized by subject groups: STEM, Social Science, Humanities, Arabic Language, and Other." (p.16)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories (single correct option from 2–5) are regionally relevant for MCQ tutoring [Q27]; the format itself is appropriate. The categorical label space (A/B/C/D/E or Arabic equivalents [Q51, Q52]) is universally usable.

OO-2: Missing categories: no jurisdiction-specific correct-answer branching for law and civics [Q27], explicitly required by deployment [elicitation A2]; no madhab-specific branching for Islamic studies [gap_id 7]; no mechanism for 'correct in country X but not in country Y' [DATASET-D8, D34].

OO-3: The Driving Test configuration encodes country-specific traffic regulations as correct answers [DATASET-D35] — a positive example of country-tied ground truth, but applied inconsistently across other subjects where country variation also exists.

OO-4: Stakeholder-driven taxonomy redesign would benefit law and civics subsets to add jurisdiction tags. This is feasible given existing per-question country metadata [Q33] but not implemented in the published label schema.

OO-5: Documented taxonomy issues harming structural validity: (a) no jurisdiction tagging in label schema for law [DATASET-D34]; (b) Jordanian-default civics presented as universal [DATASET-D1, D2]; (c) no madhab labeling [gap_id 7]; (d) null country metadata for several configurations [DATASET-D26, D27].

ID	Evidence
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q33	"In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4)
Q51	"In the case of English alphabetic output, we measure the probability of the first generated token being A, B, C, D, or E." (p.6)
Q52	"For Arabic, we measure the probability of the first generated token being @, H., h., X, or è." (p.6)
D1	What are the manifestations of change in modern Jordanian society? — country-specific Jordanian civics
D2	The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D8	— Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D26	All 5 examples: Country = null, Source = folderat.com, Subject = Islamic Studies — Islamic Studies "All" config examples have no country metadata — source appears to be generic Arabic quiz site

D27	All 5 examples: Country = null, Source = madinaharabic.com — Arabic Language General has no country metadata — sourced from online Arabic language learning site
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D35	رتم 50 دعب ىلع؟هعضو كيلىع نيأ، سكاعلا ريذحتلا ثلثم كيىلو كتبكرم ل طعت ل ا ح يف — UAE driving rule (50m warning triangle placement) — country-specific traffic regulation

Information Gaps

- Whether per-question country metadata is consistently populated across the full dataset (sample shows null for Islamic Studies All and Arabic Language General [DATASET-D26, D27])

Requires Expert Verification

- Whether subject-matter experts in law and civics deem the existing single-answer schema redeemable through metadata-stratified evaluation, or whether label-schema redesign is required

Remediation

Gap 1: Flat single-correct-answer schema [Q27] cannot represent jurisdiction-specific correctness, despite deployment requirement for country-tailored legal answers [elicitation A2].

Recommendation: Add a jurisdiction-tag overlay on top of ArabicMMLU's existing country metadata for law and civics subsets; rescore tutoring outputs against jurisdiction-conditioned ground truth rather than the flat label.

ID	Evidence
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output content (label correctness for the deployment context) is the dimension most exposed to validity violations. The 96% spot-check agreement [Q43, Q44] was conducted on 100 randomly sampled questions [Q42] without country stratification, so it cannot speak to per-country annotation quality. Annotators are exclusively from Jordan, Egypt, Lebanon, UAE, and KSA [Q31] — Morocco, Kuwait, and Palestine have no documented annotators [Q96], yet Palestine is a top-three question contributor [WEB-1] and Moroccan law content is present [DATASET-D34]. This means Palestinian, Moroccan, and Kuwaiti questions (where present) had ground-truth labels verified by workers from other national educational contexts. Empirically, the dataset analysis shows: (a) Jordanian-curriculum facts (King Talal's reign, 1952 constitution, Royal Medical Services, CBJ URL) are encoded as universally correct answers [DATASET-D3, D4, D7, D20, D21]; (b) Egypt-specific currency and institutions are encoded as universal in university economics [DATASET-D28, D39, D40]; (c) Moroccan procedural law is encoded as 'the' law answer without jurisdiction tag [DATASET-D8, D9, D10, D34]; (d) at least one likely-incorrect ground-truth label is observed [DATASET-D32]; (e) OCR-corrupted questions [DATASET-D14, D15] cannot have reliably verifiable answer keys. Pretraining contamination is also acknowledged as not ruled out [Q94, Q95]. For a deployment that requires curriculum-aligned answers per country [elicitation A1] and country-specific legal correctness [elicitation A2], the ground-truth labels do not reliably reflect regional stakeholder perspectives for Moroccan, Emirati, Kuwaiti, Saudi, Lebanese, or Palestinian cohorts.

ID	Evidence
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q42	"Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4)
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
Q44	"This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)
Q94	"It is important to emphasize that our experimental results do not provide conclusive answers regarding the performance of LLMs in Arabic." (p.9)
Q95	"to a lack of sufficient information about its training regimen. As such, we cannot assert that the model's pretraining data is free from contamination." (p.10)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
D3	"ب وه ةفأك ةل وذل نوؤش ىلع ةماعل ةيالول بحاص سلجلم" with answer "عرازول سلجلم" — "The council with general sovereignty over all state affairs is the Council of Ministers" — specifically Jordanian constitutional structure
D4	"Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom
D7	0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact
D8	— "ىوعدلا رهوج يف مكحل رودص دعب الإ عوفد وأ ضراع عازن يف ةرداصل وأ ةيديهمتمل ماكحلأ فانئتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure

D9	عافدل اب ةفل كملم ةيموكحل اطلسل او لدعل ربزو نم رارب...نيكي كدلل ةيئاضقلا ططرشل ا طباض ةقص لي وخت نكمي Moroccan judicial police powers — references "يكلملا كدلا" (Royal Gendarmerie of Morocco) — ينطولا
D10	Moroccan civil party withdrawal in criminal proceedings — عيحص ةبوجل ا عيجم: مكحل رودص لبقي ندم فرطلا لزانن اذ
D14	Severely OCR-distorted text about authority distribution in organizations, from Egyptian source — لري تدت...ةتفل تخسل ا تيرادل ا تايهت تدسل او صاختل ال ني تب ةطلتدل ا عتييهت ةتجرد OCR-distorted text about authority distribution in organizations, from Egyptian source
D15	Severely OCR-corrupted management question with unreadable answer options — ت ح... ت محرخل... ا مدختسل ا ةسئاد ططخل ا عل ل اثم دعي ل unreadable answer options
D20	"Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services — ةي بطلا ةني دمل ا يفشتسم :اهنم...تاسسؤملا نم ديدعل ا ةيموكحل ا ةيكلمل ا ةي بطلا تامدخلا لمشت "Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school — ةنس همكح ةدمو 1952م: يفل كل ذو ةيحص بابسل ا هتالو تهتن ني سحل نب لال طلكلم "King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school
D28	Egyptian currency (جني = 100 Egyptian pound) in university economics — Egyt-specific monetary context — 10% :وه ةرمثتسملا لاول امل ا عل ةئاعلا لدعم...هينج 1000 بولطم رامثتسال ا مجح ناك اذ Egyptian currency (جني = 100 Egyptian pound) in university economics — Egyt-specific monetary context
D32	تايمكب هرفاوت: ل ا كل ذ ي ف ببسل ا عجي...ي عي بطلا زاغلا و طفنلا نم ةريكب تاوثرثب ةيبرعل ا نبال ا عل هلل ا عان "Gas is available in limited quantities" — possibly incorrect answer key for general Arab world social science
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)

Strengths

- Annotator credentials are documented: Master's students/Research Assistants and Bachelor's-degree-holders, all native MSA speakers [Q35, Q39]
- 96% inter-annotator agreement on 100-question spot check provides a baseline quality signal [Q43]
- Compensation exceeded monthly average wages in respective countries [Q36]
- One-hour orientation workshop conducted before data collection [Q40]
- STEM ground truth (math, biology, physics, factual CS) is curriculum-neutral and reliable across all eight countries [DATASET-D24, strength 5]

ID	Evidence
Q35	"Our internal workers are Master's students and Research Assistants in Computer Science, while the external workers hold Bachelor's degrees." (p.4)
Q36	"We ensured competitive compensation for the workers, exceeding the monthly average wage in each respective country." (p.4)
Q39	"While our workers are native speakers of Modern Standard Arabic with at least Bachelor's degrees, we maintain the quality of our dataset construction through meticulous steps." (p.4)
Q40	"Firstly, we conducted a 1-hour workshop before the data collection stage to clarify the process." (p.4)
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
D24	ا ا هت ي ب ث ت م ت ي تال (CO2) تائ ي ز ج ددع مكف ن فل اك ةقلح نع زوكولغ تائ ي ز ج (3) جتن اذ Palestine-sourced biology; universal factual content

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?

OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels reflect regional stakeholder perspectives unevenly. For STEM and curriculum-neutral content, labels are reliable across all eight countries. For civics, history, social science, economics, and law, labels reflect the originating exam's jurisdiction without country-stratified validation [DATASET-D1, D7, D34].

OC-2: Disagreement is highly likely between original annotators and regional populations for: (a) Moroccan students reviewing Jordanian civics [DATASET-D1, D2]; (b) UAE/Kuwait students reviewing Egyptian university economics [DATASET-D28]; (c) Egyptian/Saudi/Jordanian/UAE/Kuwaiti/Lebanese/Palestinian law students reviewing Moroccan procedural law [DATASET-D8, D9, D10]; (d) Palestinian students reviewing Jordanian-narrative King Abdullah I content [DATASET-D31].

OC-3: Annotator demographics documented [Q31] but limited: 10 annotators total, all from Jordan/Egypt/Lebanon/UAE/KSA. No Morocco/Kuwait/Palestine representation [Q96]. No formal datasheet or data statement beyond paper text.

OC-4: Re-annotation by representative annotators from Morocco, Kuwait, and Palestine is essential before deployment use, especially for civics, history, law, social science, and economics subsets.

OC-5: Aggregation: 96% agreement reported as a single global figure [Q43] without country stratification — minority-country perspectives are erased in the aggregate. No formal IAA statistic beyond this [Q43].

OC-6: Documented label issues: (a) Jordanian-curriculum facts as universal ground truth [DATASET-D1, D3, D4, D7, D20, D21]; (b) Egyptian university content as universal [DATASET-D28, D39, D40]; (c) Moroccan-only law ground truth [DATASET-D34]; (d) no Palestinian annotator review of Palestinian content [Q96, WEB-1]; (e) at least one likely-incorrect answer key observed [DATASET-D32]; (f) OCR-corrupted questions where verifiability is compromised [DATASET-D14, D15]; (g) potential pretraining contamination not ruled out [Q94, Q95].

ID	Evidence
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
Q44	"This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)
Q94	"It is important to emphasize that our experimental results do not provide conclusive answers regarding the performance of LLMs in Arabic." (p.9)
Q95	"to a lack of sufficient information about its training regimen. As such, we cannot assert that the model's pretraining data is free from contamination." (p.10)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
D1	What are the manifestations of change in modern Jordanian society? — "What are the manifestations of change in modern Jordanian civics"
D2	The first coalition government in Jordan in 1952 was known as the ___ government — "The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D3	The council with general sovereignty over all state affairs is the Council of Ministers — "The council with general sovereignty over all state affairs is the Council of Ministers" — specifically Jordanian constitutional structure
D4	Rights under the [Jordanian] Kingdom's constitution — "Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom
D7	0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact

D8	— يوعدل رهوج يف مكحل رودص دعب ال عوفد وأ ضراع عازن يف ةرداصل وأ ةيدي مهمتلا ماكحلأ فانئتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D9	عافدل اب ةفكلمل ةيموكحل اطلسل او لدعل ربزو نم رارقب...ني ك ردلل ةيئاضقلا ططرشل طباض ةفص لي وخت نكمي عافدل اب ةفكلمل اكردل" (Royal Gendarmerie of Morocco) — Moroccan judicial police powers — references
D10	Moroccan civil party withdrawal in criminal proceedings — ةحي حص ةبوجأل عي مج :مكحل رودص لبقي ندمل فرطل ل زانت اذا
D14	Severely OCR-distorted text about authority distribution in organizations, from Egyptian source — ىل يري تذت...ةتفل تخسل ةتيرادلإا تايهت تدسل او صاختلأا ني تب ةطلتدل عتيهت ةتجرد
D15	Severely OCR-corrupted management question with unreadable answer options — ت... حل... مخرخل...مادختسلإا ةسئاد ططخل ىل عل ءل اثم دعي لا
D20	"Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services — ةي بطل ةني دمل ىف شتسم :اهنم...تاسس ؤمل نم ديدعل ةيموكحل ةيكلمل ةي بطل تامدخل لمشت
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school — ةنس همكح ةدمو ، 1952م :في كل ذو ةي حص بابسأل هتيلو تهنتا نيسحل نب لال طلكمل
D28	Egyptian currency (ج = ١٠٠٠٠٠ Egyptian pound) in university economics — Egypt-specific monetary context — وه ةرمثتسم ل لاومألا ىل ع دئاعل لدعم...هي ن ج ١٠٠٠٠٠ بولطم رامثتسال م ج ج ناك اذا
D31	King Abdullah I's positions on Zionism — highly sensitive political content, Jordanian framing of Palestinian history — ني طسلف في اعامطأو ةينوي هصلال ةكرحل نم لوأل هلل ادبع كلمل فق او م نم تسيل ةيتالآا نم ةدحاو
D32	تاي مكب هرفاوت :ىل ك ل ذ في ببسل ع جري...ي عي بطلا زاغل او طفلن نم ةري بك تاو رثب ةي برعل اندال ب ىل عل هلل مع أن "Gas is available in limited quantities" — possibly incorrect answer key for general Arab world social science — ةلي ق
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)

Information Gaps

- Country-stratified inter-annotator agreement is not reported [Q43]
- Pretraining contamination extent is not measured [Q95]
- Total proportion of incorrect answer keys beyond the 4% noise floor implied by 96% agreement is unknown

Requires Expert Verification

- Per-country curriculum experts (especially Moroccan Maliki Islamic studies, Kuwaiti civics, Palestinian national history, UAE 4-4-4 curriculum) to re-validate ground-truth labels
- Legal experts from Egypt, UAE, KSA, Jordan, Lebanon, Kuwait, Palestine to assess law-subset jurisdictional correctness

Remediation

Gap 1: Ground-truth labels for Palestinian, Moroccan, and Kuwaiti questions were verified by workers without those national curriculum backgrounds [Q31, Q96]; 96% agreement is a global figure not stratified by country [Q43].

Recommendation: Recruit at least 2–3 curriculum experts each from Morocco, Kuwait, and Palestine to re-verify ground-truth labels for at least the civics, history, social science, and law subsets relevant to those countries. Report country-stratified IAA before deployment use.

ID	Evidence
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)

Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is the strongest dimension for this deployment. Both the benchmark and deployment use single-token MCQ accuracy [\[Q50, Q53\]](#), and the elicitation explicitly confirmed that for MCQ tasks 'explanations are not graded, so any correct explanation adequately supports the student's understanding without exam risk' [elicitation A3]. The benchmark provides multiple prompt-language and output-alphabet configurations [\[Q47\]](#) with detailed per-configuration results [\[Q105, Q106, Q108, Q110\]](#), offering rich diagnostic capacity. Calibration analysis ($r > 0.9$ between answer probability and accuracy [\[Q80\]](#)) provides additional signal quality. Models are evaluated in zero-shot and few-shot settings across 35 models [\[Q45, Q46\]](#), giving comparative reference points. The remaining concerns are residual: the form does not capture explanation quality (which is acceptable per elicitation A3) and does not evaluate generative tutoring output that the deployment may also produce. Literacy is a non-issue since target users are enrolled students [regional_context literacy section].

ID	Evidence
Q45	"Our experiments focus on zero-shot and few-shot settings across 35 models." (p.5)
Q46	"This includes 22 open-source multilingual models (XGLM (Lin et al., 2022), BLOOMZ (Muennighoff et al., 2022), mT0 (Muennighoff et al., 2022), Falcon (Penedo et al., 2023), and LLaMA2 (Touvron et al., 2023), across various sizes), 11 open-source Arabic-centric models (AraT5 (Nagoudi et al., 2022), AraGPT2 (Antoun et al., 2021b), AceGPT (Huang et al., 2023) and Jais (Sengupta et al., 2023), also across various sizes), and 2 closed-source models (GPT-3.5: gpt-3.5-turbo (Ouyang et al., 2022) and GPT-4: gpt-4-0613 (OpenAI, 2023))." (p.5)
Q47	"We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output." (p.5)
Q50	"Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options." (p.6)
Q53	"For closed-source models, we determine the answer based on the first token generated in the text using a regular expression." (p.6)
Q80	"In Figure 8, we observe that the three open-source models are well calibrated with correlation scores $r > 0.9$." (p.8)
Q105	"Zero-shot LLM performance (% accuracy) with Arabic prompt and Arabic alphabetic output, combined across subject groups." (p.16)
Q106	"Average" means the average across all questions in ArabicMMLU." (p.16)
Q108	"Zero-shot LLM performance (% accuracy) with Arabic prompt and English alphabetic output, combined across subject groups." (p.17)
Q110	"Zero-shot LLM performance (% accuracy) with English prompt and Arabic alphabetic output, combined across subject groups." (p.18)

Strengths

- Single-token MCQ accuracy directly matches deployment's MCQ scoring needs [\[Q50, Q53\]](#)
- Four prompt-language/output-alphabet configurations tested, with per-configuration results published [\[Q47, Q105, Q106, Q108, Q110\]](#)
- Calibration analysis ($r > 0.9$) confirms answer-probability reliability for open-source models [\[Q80\]](#)
- Per-subject and per-level result breakdowns enable diagnostic targeting [\[Q102, Q103\]](#)

ID	Evidence
Q47	"We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output." (p.5)
Q50	"Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options." (p.6)
Q53	"For closed-source models, we determine the answer based on the first token generated in the text using a regular expression." (p.6)
Q80	"In Figure 8, we observe that the three open-source models are well calibrated with correlation scores $r > 0.9$." (p.8)
Q102	"Table 8 presents the detailed zero-shot results across subjects and education levels, while Table 9, Table 10, Table 11 display the results with different prompts and alphabetic outputs (complementing the main result at Table 8)." (p.15)
Q103	"Zero-shot LLM performance (% accuracy) with English prompt and English alphabetic output, for each subject and education level." (p.15)
Q105	"Zero-shot LLM performance (% accuracy) with Arabic prompt and Arabic alphabetic output, combined across subject groups." (p.16)
Q106	"Average" means the average across all questions in ArabicMMLU." (p.16)
Q108	"Zero-shot LLM performance (% accuracy) with Arabic prompt and English alphabetic output, combined across subject groups." (p.17)
Q110	"Zero-shot LLM performance (% accuracy) with English prompt and Arabic alphabetic output, combined across subject groups." (p.18)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Expected output modality (single-letter MCQ answer) matches the deployment's primary MCQ tutoring use case [[Q50](#), [Q53](#); [elicitation A3](#)]. The deployment may also need free-form explanations, which the benchmark does not evaluate — but the elicitation confirmed this is non-critical for MCQ contexts [[elicitation A3](#)].

OF-2: Text-to-speech is not required by the deployment (text-based interface per `regional_context` platform_context). Not applicable.

OF-3: Target population is enrolled students with high literacy by definition [`literacy_and_educational_attainment` note]; mobile-dominant access is documented [[WEB-11](#)]. Output form is compatible with regional access patterns.

OF-4: Documented form mismatches are minimal: (a) explanation-quality output is not evaluated, but elicitation confirmed this is acceptable [[elicitation A3](#)]; (b) generative tutoring output (e.g., step-by-step explanations) falls outside benchmark scope but is a deployment use case the benchmark does not validate.

ID	Evidence
Q45	"Our experiments focus on zero-shot and few-shot settings across 35 models." (p.5)
Q50	"Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options." (p.6)
Q53	"For closed-source models, we determine the answer based on the first token generated in the text using a regular expression." (p.6)
WEB-8	~100% internet penetration in UAE/KSA/Kuwait — text-based output reaches target users (https://statisticsoftheworld.com/ranking/internet-users)

WEB-11

83% mobile ownership in Arab States; mobile-dominant text delivery is feasible
(https://www.itu.int/dms_pub/itu-d/opb/ind/D-IND-SDDT_ARB-2025-PDF-E.pdf)